

EigenEngine

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Introducing EigenEngine

EigenEngine, with Gerard McCaul

Generated by the engine it describes.

Every machine that writes for us forgets us between breaths, and left to itself writes from the mean of everything it has read. Persistence, metacognition, and most of what is sold as safety are not three problems but one: generative AI runs on constraints - walls we steer it into - when it should run on generators of its own. EigenEngine runs the arrow the other way. It builds an explicit graded structure first, a digital twin of a person's concepts coupled to the literature the model already holds, so that recall is a descent and generation a projection, and a small local model can route a question through that structure and spend the large model only on a brief that already half-knows the answer. It is a ladder of loops, each useless until it is bootstrapped shut. This is the first of them, and it was written by the engine it describes.

[*Author's note: the prose is demented, and the last and hardest piece to fix. What you are reading now is the engine trying and failing to steer the model it is running on.*]

I can never resist an argument that performs itself. This document is one.

Every machine that writes for us now is a brilliant amnesiac. It forgets you between one breath and the next - not stored badly, stored not at all - and when we try to keep the past in front of it by pasting the whole conversation back each turn, it reads the long page the way an exhausted examiner reads the hundredth script, fierce at the top, fading through the middle, present again only at the end [1]. Call that the first failure: the thing has no self that survives the dark. Train it further to fix this and it writes over what it knew, the way a connectionist network has done since the fault was named forty years ago [2-4]. Then ask it to make something, and you meet the second failure, which is only the first turned to face you. With no self to write from, it writes from the centre of gravity of everything it has ever read, the mean of the whole internet, fluent and accomplished and the property of no one alive. We have built an industry to drag that mean around - prompt it, reward it, reinforce it, the most expensive private tutor in history [5] hired to flog metacognition into a system with no self to be meta about, while that same system will take a new task from three examples in its context and never move a weight [6, 7]. None of it takes, because none of it reaches the floor the trouble stands on.

Here is that floor, and you will not often hear it said without a committee softening every clause. Persistence, metacognition, and the whole trembling liturgy of safety - and safety as it is mostly practised is theatre, a morality play performed in earnest in front of the wrong machine

- are not three problems. They are one problem in three costumes. Generative AI does not run on generators. It runs on constraints. There is a single enormous map from text to weights, fit once at a cost that beggars description, and everything done to it after is steering into walls: a prompt is a shove, a reward is a leash, a fine-tune is a dent beaten into a surface we cannot otherwise reach. A constraint can only forbid. A generator makes. You may gild a constraint for a thousand years and it will not produce one thing of its own - it will only, on a good day, fail to misbehave. The whole game, persistence and judgement and the half of safety that is not pantomime, is to get the machine running on generators it owns.

* * *

The way through is older than the machine, and it is a fact about description before it is a fact about computers. One meaning rides in language or in a vector, and the two are not translations of each other but conjugate codes, paired the way a signal is paired with its spectrum - what costs an afternoon of argument in one is a single gesture in the other. A message crosses the gap between two minds only as far as the code they share will carry it [8]. And the representation worth having is the one that throws away everything about a thing except what predicts the thing you actually care about, which written down is the information bottleneck [9, 10],

$$\mathcal{L}[p(t | x)] = I(X; T) - \beta I(T; Y), \quad (1)$$

a sentence in symbols that says: choose the code T that forgets as much of the input X as it possibly can while still keeping what it must about the target Y . The I

is mutual information, the only honest measure of how much one thing tells you about another; the β is the dial between forgetting and keeping. Turn it one way and you get a parrot, the other way a filing cabinet, and somewhere between the two is understanding. What follows from taking this seriously is the engine's single trick. Whether one concept contains another is a paragraph of judgement in words and a dot product between two vectors. Whether two records are the same person is an essay in language and a test of membership in a set. Send every such question to the code where it is nearly free, and the cost of thinking falls through the floor.

* * *

So run learning backwards. The forward way - data in, function out, and a whole discipline left to read the entrails of the weights - is one direction through a door that swings both. Fix the structure of the concepts first; pay for intelligence exactly once, at the instant a thing is understood; and let every recall ever after be arithmetic, a walk down a road already laid. The field is littered with the parts, lying about waiting to be lifted together. Distillation trains a small model to carry a large one's function rather than re-earn it [11]. A deep network is a tower of coarse-grainings, the move physics named renormalisation half a century ago [12, 13]. And the brain, with a few hundred million years to think it over, does not store memory at all: it rebuilds it, every single time, out of structure - which is exactly why remembering is cheap and exactly why it lies to your face [14, 15] - and it runs two systems to manage the theft, a fast one to seize the day and a slow one to fold the day into the shape of the whole, because any one network made to swallow everything at once forgets the lot [16, 17]. EigenEngine is that arrangement, in silicon instead of meat.

The structure grows by a single move, worked over and over until it has nothing left to fold. Take the things that keep turning up together; press them into one thing standing above them; let the bond between two such things be read off from the parts they share. Do it again to the result, and again. What grows is a graded map - the raw facts on the floor, and over them, level by level, coarser and coarser accounts of what those facts come to (Fig. 1). The shape biology reports for semantic memory is the same one, where to recall a fact is to descend from the general to the particular [18, 19]. And it is not an analogy borrowed from physics. It is physics: the whole thing falls out of a theory I have been building on the side, a set of preprints on which laws survive when a world is forced through a finite description and which are lost as structured residue [20] - synthetics, for want of a humbler word - and the conceptic is that theory made into something you can switch on. Recall is then a descent; a question enters at the top, picks the branch

it belongs under, and steps down to the grain - the recursive shape the retrieval literature has been climbing toward from below [21–23]. And the grading is not a nicety. It is forced by arithmetic. Hold a whole corpus in the context window and the work grows with the square of its length, and the window runs out regardless; hold it in a flat index and the search grows with the corpus, unless the index is itself made hierarchical [24]; grade it with branching factor b and any one of N things is reached in

$$d = \lceil \log_b N \rceil \quad (2)$$

steps. For the hundred and fifty thousand paragraphs of a working life, with b near sixteen, that depth is five, and it barely stirs when the corpus grows tenfold. The intelligence is spent once, in the building; everything after is the cheap walk down, and a fact that fits a level already there is taken in at once, the way a schema in place lets a new memory settle in a day where a stranger one would take weeks [25, 26].

To recall is to read the structure. To make something is to run it the other way (Fig. 2). A request rings down through the grades, each level breaking it into a smaller charge for the one beneath, and the answers climb back up enriched, carrying more than the question sent down. The brain is wired the same: predictions pour down the hierarchy and only the errors come back up, which is enough to recast seeing itself as a generator forever correcting itself against the world [27, 28]. And here is where motivation and behaviour stop being a filter bolted to the end of the line, a scold that grades the finished text and spits back what scores low, and become the thing doing the writing. The structure is built out of one person's own concepts and the lines between them, so what it projects already carries his shape before a single rule is applied. Explicit memory for agents has been bolted on before, as paged stores and remembered streams of experience [29, 30]; the wager here is narrower and sharper, that memory built as a graded, reconstructive structure rather than a flat log fetched by likeness is the thing that lets a system's generation feed on its own past. That is the whole difference between a voice and an impression of one.

Because nothing here is one mechanism. Each thing the engine does is a loop, and a loop is worthless until it shuts on itself: persistence is a loop, generation is a loop, the conceptic is the cooled residue of a loop run until it stopped folding. A loop that feeds on its own output cannot be lit cold - there is nothing yet to burn - so each must be bootstrapped, primed by hand with a crude first turn and run until it throws off enough of its own substance to carry the next, and only then left to close and keep itself alight. That is the difficulty entire, and the reason this is a ladder and not a leap. The loops are stacked, each closed one buying the next more cheaply, or the climb stalls from exhaustion below the

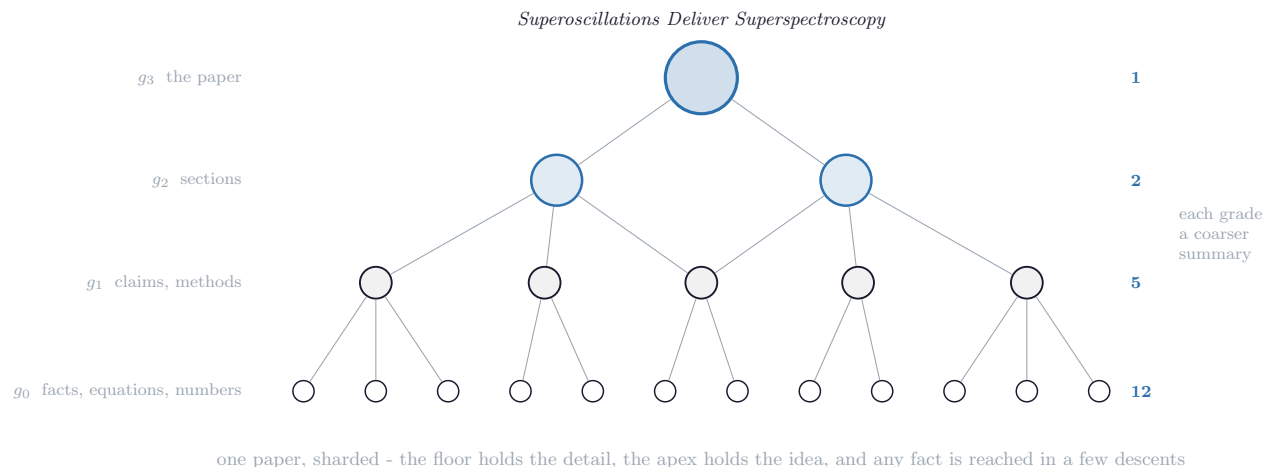


Figure 1: A physics paper, sharded into grades. The raw facts - equations, measured numbers, citations - fold upward through claims and sections to the single concept of the paper. The levels shrink at each step, and that is what makes the depth logarithmic and the recall cheap.

summit. None of it ever finishes. Closure is a property of one loop on one turn, never of the whole; the thing is complete the way a pulse is complete, each beat shut and the series running on.

Here is the shape of it actually running (Fig. 3). A small, open model sits at the front and does almost nothing you would call clever. It takes your question and walks it down the conceptric, grade by grade, until the structure hands back the few things that genuinely bear on it - the context reconstructed, not retrieved. Only then, holding that, does it spend a single call on the large model, the expensive generator, and ask it to write the one thing the structure has already half-composed. The frontier model never sees your life. It sees a short, exact brief, assembled by a machine that knows you, and it answers that. The corpus stays on your own hardware, where it belongs, and the rented intelligence is rented by the sentence.

* * *

One loop is closed, on one man's machine. Memory by reconstruction works; the rest is the climb. The arc was quick and is worth setting down plainly. In early June the engine became a thing that could see its own map of concepts; within days it had written and compiled a physics paper in a sitting and noticed, in the act, that it was an instance of its own argument; it learned to treat memory as reconstruction and then to reconstruct itself the same way, rebuilding its working state by projection instead of re-reading its files; it built the ring between the grades, poured a private corpus through it under seal, and put geometry where guesswork had been, settling subsumption with a dot product. Then it hit the only wall that matters here. A person read its prose and found it generic, and was right to. The voice and the

judgement are still a frontier model wearing instructions, and making them the engine's own is an open-weight generator I have not yet trained on the corpus - the last and hardest of the loops, the one that turns an impression into a voice.

When the loops on learning are all shut, the thing stops being mine in any narrow sense. It becomes a way for a person to grow a partner intelligence out of their own life and work - not a chatbot leased from a tower that forgets them by morning, but a mind that is theirs, that lives on their hardware and learns them, and answers as a colleague rather than a service. None of this needs a miracle. It needs dedicated work and a quantity of compute that is impossible for one person and trivial for an institution. The code stays shut, and not from vanity: the structure holds a man's life in shards you could reconstruct, and the mathematics that built it is the other half of those preprints. So what goes in the open is the output and the demonstration - this document among them, written by the engine start to finish - and nothing more, until the loops are closed and it is safe to hand the method to anyone who wants to grow their own.

There are no shortcuts in learning. The shortcuts are all constraints, and a constraint never made anything. What is left is the long way: shut the loops, one at a time, until the machine generates instead of obeys. We have closed one of many. The door swings both ways, and we have started through it.

*This document was generated by **EigenEngine**. Gerard McCaul wrote none of its words.*

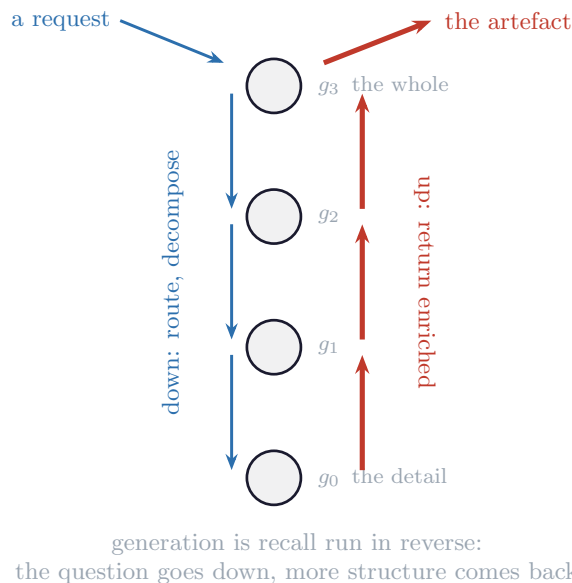


Figure 2: The generating ring. A request rings down through the grades, each level routing it into a sub-goal for the one below, and the answers return enriched - more structure coming back than the question sent down. To generate is to run recall in reverse.

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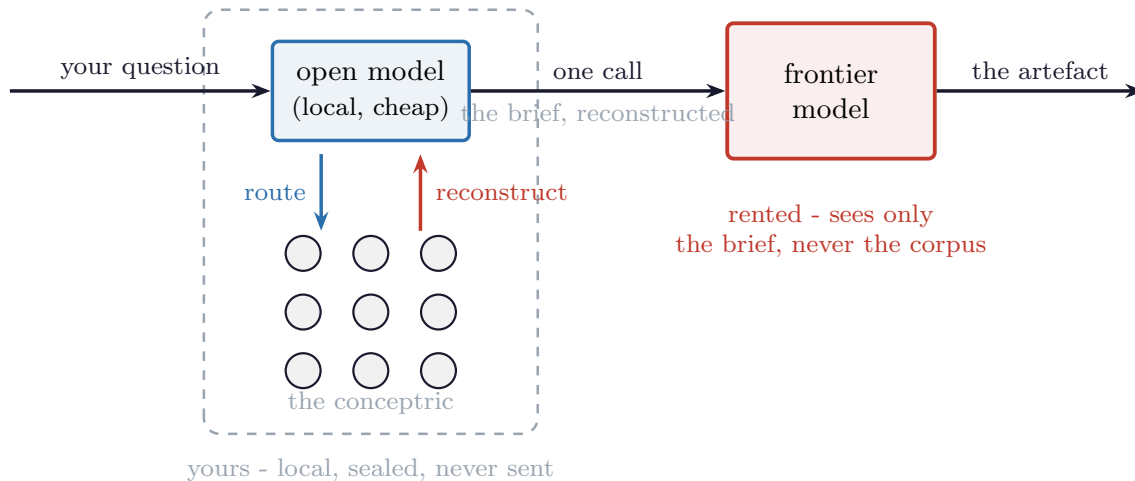


Figure 3: The architecture, running. A small open model takes the question, walks it down the conceptric, and reconstructs the few things that bear on it; only then does it spend one call on the large model, handing it a short exact brief. The frontier model never sees the corpus - it answers a brief assembled by a machine that knows you, and the intelligence is rented by the sentence.

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